

Acute Dizziness & Vertigo — Clinical Reference

Sort by **Timing + Triggers** (modified TiTrATE), not by symptom quality. Pick the matched bedside test. Adapted from TeachIM (Melton, 2021) · 2023 GRACE-3 · CC BY-NC 4.0.

Step 1 — Categorize (Timing & Trigger)		
Category	Duration	Test → lead dx
Episodic, positional	sec–min	Dix-Hallpike → BPPV; orthostatics
Spontaneous episodic	min–hr	Hx, Weber/Rinne → vestib migraine, Meniere, vasovagal/panic, TIA
Continuous, trauma/toxin	persistent	Med review, otoscopy → drug-induced, barotrauma
Continuous, spontaneous (AVS)	hr–days	HINTS → neuritis vs stroke

- Step 1b — History on everyone
- CV risk: HTN, DM, AF, CAD, smoking (↑ posterior-circ stroke)
 - Hearing loss / tinnitus (Meniere, labyrinthitis)
 - Focal neuro sx, new headache/neck pain
 - Meds/changes: AEDs, sedatives, antiHTN, analgesics
 - Substance use; recent trauma/barotrauma

- Step 2 — Dix-Hallpike (positional)
- Head 45° to side, lay supine, head 20° off table; watch 30 s; repeat other side
 - **Positive** = transient upbeat-torsional nystagmus + vertigo = **pc-BPPV**
 - Sens ~79% / Spec ~75% — better at ruling IN
 - **Treat now:** Epley canalith-repositioning (strong rec, GRACE-3). Half-somersault = home option, less effective. Vestibular PT if refractory.
 - **Negative** → orthostatics (SBP ↓≥20 / DBP ↓≥10 within 3 min), reconsider

Step 3 — HINTS (continuous AVS only)		
Use ONLY in continuous, spontaneous vertigo with nystagmus , by a trained examiner.		
Component	Peripheral (OK)	Central (bad)
Head Impulse	Catch-up saccade (abnl VOR)	Normal / intact (no saccade)
Nystagmus	Horizontal, unidirectional	Bidirectional or vertical
Skew	None	Vertical skew (cover test)

Rule: **ANY central finding → stroke.** ALL reassuring → vestibular neuritis. Mnemonic **INFARCT** = Impulse Normal, Fast-phase Alternating, Refixation on Cover Test.

- Management by diagnosis
- **BPPV:** Epley at bedside; half-somersault home; PT if refractory. No imaging.
 - **Vestibular neuritis:** antiemetics + short-course vestibular suppressant/anticholinergic (limit to days — chronic use blocks central compensation). Steroids only via shared decision-making if ≤3 days (weak evidence).
 - **Central / stroke pattern:** activate stroke pathway; MRI-DWI + MRA; ASA + statin; tPA/thrombectomy per window & resources.
 - **Acute idiopathic SNHL:** steroids ARE indicated.

Do / AVOID (2023 GRACE-3)	
DO	AVOID
Epley for +Dix-Hallpike (strong)	Routine noncontrast head CT to r/o posterior stroke (insensitive; strong rec against)
HINTS only in AVS w/ nystagmus, trained exam	HINTS in pts w/o nystagmus or positional-only sx (inaccurate)
MRI-DWI/MRA if central suspected	False reassurance from early negative MRI (misses ~15–20% posterior stroke <48 h)

- Red flags → central work-up
- Normal head impulse **WITH** continuous vertigo
 - Direction-changing or vertical/torsional nystagmus; skew
 - Truncal ataxia — can't sit/stand unaided
 - New headache / neck pain; any focal deficit
 - Vascular risk factors + sudden persistent vertigo